

ANTIGRAVITY — HEALTH PORTABILITY ANALYZER

TECHNICAL ARCHITECTURE & IMPLEMENTATION GUIDE (A to Z)

Built for Prudential Health India | Generated on 23 Jun 2026

1. Executive Summary & Purpose

AntiGravity is an AI-powered portal designed to streamline **health insurance portability** into Prudential Health India. In India, policyholders are legally entitled by the **IRDAI** (Insurance Regulatory and Development Authority of India) to port their existing health insurance policies from other insurers (e.g. Star Health, Care Health, Niva Bupa) to Prudential, retaining waiting period credits for pre-existing diseases (PEDs).

Historically, this process has been highly manual, requiring agents to inspect lengthy policy wordings and manually fill out statutory forms. AntiGravity automates this end-to-end pipeline by:

- **Parsing competitor PDFs** (digital or scanned) to extract 25+ insurance data fields.
- Performing **IRDAI portability eligibility checks** (such as the 45-day pre-expiry timeline).
- Evaluating **coverage gaps** (co-payments, room rent limits) and recommending the optimal Prudential plan.
- **Auto-filling the IRDAI Annexure A portability form** as a pre-filled PDF ready for signing.

2. System Architecture & Multi-Agent Design

AntiGravity is architected around the **Google ADK (Agent Development Kit)**, leveraging a hierarchical multi-agent structure. This decouples responsibilities, allowing individual sub-agents to focus on extraction, analysis, and recommendations independently.

Agent / Layer	Role & Responsibility	Key Integration Tools
Orchestrator Agent (Root Agent)	Coordinates the execution flow. Receives the input PDF and invokes sub-agents sequentially.	google-adk Runner
Extraction Agent (Sub-agent)	Responsible for extracting policy text and structuring it into a 25+ field schema.	pdf_extractor.py, llm_parser.py
Analysis Agent (Sub-agent)	Performs regulatory eligibility validation (IRDAI rules) and analyzes competitor coverage gaps.	rules/settings.py
Recommendation Agent (Sub-agent)	Matches the customer profile to the optimal Prudential plan and triggers report/form generation.	plan_matcher.py, report_generator.py, form_filler.py

ANTIGRAVITY — HEALTH PORTABILITY ANALYZER

TECHNICAL ARCHITECTURE & IMPLEMENTATION GUIDE (A to Z)

Built for Prudential Health India | Generated on 23 Jun 2026

3. Under-the-Hood Technical Components (A to Z)

A. PDF Text Extractor (pdf_extractor.py)

Extracting text from insurance policies can be difficult because they are often structured in complex grids or scanned as images. The PDF Extractor uses a 3-layer extraction strategy:

- pdfplumber**: Best for digital PDFs containing tables. Extracts text and joins table rows logically.
- PyPDF2**: Runs as a secondary fallback if `pdfplumber` fails.
- pytesseract OCR**: Runs as a final fallback for scanned PDFs. Converts pages to images via `pdf2image` and runs OCR.

B. Dual-Mode LLM & Regex Parser (llm_parser.py)

To ensure high flexibility and 100% uptime, the parser supports two modes:

- LLM Mode (Claude / Gemini)**: Submits policy text to the LLM to extract structured JSON data according to a strict Pydantic-style schema.
- Rule-Based Offline Mode**: If LLM keys are absent, this mode runs regex and template matching to extract values locally. This guarantees the demo works without API dependencies.

C. Plan Matcher Engine (plan_matcher.py)

Scores Prudential plans against the customer's current policy. It compares:

- Base Sum Insured Mapping**: Ensures sum insured matches or exceeds current coverage.
- Pre-Existing Disease (PED) Credit**: Calculates credit for continuous years covered, reducing wait times.
- Financial Savings**: Highlights the exact annual savings compared to their current premium.

D. Report Generator (report_generator.py)

Generates two formats: a raw structured **JSON report** (easy to consume via APIs) and a beautifully styled **HTML report** (optimized for agent printing or sharing with clients via WhatsApp/Email).

E. IRDAI Form Auto-Filler (form_filler.py)

The Annexure A portability application is a regulatory requirement. Using `ReportLab`, the tool renders a clean, professional, two-page A4 application form pre-filled with customer details, existing insurer history, and the recommended Prudential plan.

4. API & Integration Layer (FastAPI)

The system exposes a FastAPI REST API running on port `8000` to support integrations with frontend portals or mobile apps:

Method & Route	Description	Expected Output
POST /analyse	Uploads PDF file or takes PDF URL parameter.	JSON with comparison and generated file links
GET /reports	Lists all generated portability reports.	List of report metadata & summaries
GET /report/{id}/html	Fetches the styled HTML summary report.	HTML FileResponse (viewable in browser)
GET /report/{id}/form	Downloads the pre-filled Annexure A PDF.	PDF FileResponse (ready to download & sign)

ANTIGRAVITY — HEALTH PORTABILITY ANALYZER

TECHNICAL ARCHITECTURE & IMPLEMENTATION GUIDE (A to Z)

Built for Prudential Health India | Generated on 23 Jun 2026

5. Setup, Verification & Testing

The project is fully covered by unit tests validating all parts of the pipeline.

A. Dependency Setup

Install all core dependencies via requirements.txt:

```
pip install -r requirements.txt
```

B. Running Unit Tests

Run the pytest suite to verify all extractors, matchers, and form filling functions:

```
python -m pytest tests/ -v
```

C. Launching API & Frontend

Start the FastAPI server:

```
python api.py
```

Open your browser and navigate to: **http://localhost:8000/app/** to interact with the frontend.

Summary of verified components:

- ✓ Multi-agent orchestration via Google ADK verified.
- ✓ 21/21 Unit tests passed.
- ✓ PDF text extraction for Apex Policy and Portability Kit validated.
- ✓ ReportLab Annexure A PDF generator active.
- ✓ FastAPI REST endpoints running and fully integrated.